

PlantVillageVQA: A Visual Question Answering Dataset for Benchmarking Vision-Language Models in Plant Science

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Abstract

PlantVillageVQA is a large-scale visual question answering (VQA) dataset derived from the widely used PlantVillage image corpus. It was designed to advance the development and evaluation of vision-language models for agricultural decision-making and analysis. The PlantVillageVQA dataset comprises 193,609 high-quality question answer (QA) pairs grounded over 55,448 images spanning 14 crop species and 38 disease conditions. Questions are organized into 3 levels of cognitive complexity and 9 distinct categories. Each question category was phrased manually following expert guidance and generated via an automated two-stage pipeline: (1) template-based QA synthesis from image metadata; and (2) multi-stage linguistic re-engineering. The dataset was iteratively reviewed by domain experts for scientific accuracy and relevancy. The final dataset was evaluated using three state-of-the-art models for quality assessment. Our objective remains to provide a publicly available, standardized and expert-verified database to enhance diagnostic accuracy for plant disease identifications and advance scientific research in the agricultural domain. Our dataset will be open-sourced at <https://huggingface.co/datasets/SyedNazmusSakib/PlantVillageVQA>.

1 Background & Summary

Plant diseases threaten global food security and farm productivity. Research indicates that plant pests and diseases are responsible for the loss of as much as 30% of global food crop yields annually [3]. This results in famine, malnutrition, and food insecurity for hundreds of millions of people worldwide. In most cases pest and fungal invasions spread rapidly due to diagnostic delay. Therefore, precision tools are now needed for early symptoms detection and targeted interventions.

Machine learning has advanced plant pathology by enabling automated identification of disease symptoms. Convolutional neural networks and transformer models can reliably detect

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leaf diseases in various crop species [4, 5, 1, 2]. However, most existing frameworks focus only on classification and do not provide insights into symptom causation or context.

Visual Question Answering (VQA) [6, 9] combines image understanding with natural language processing to answer queries about visual content. VQA databases go beyond classification by allowing interactive question–answering [12]. This allows trained models to capture complex relationships in the images. As such, the application of VQA now spans multiple domains. These include: educational tools [15, 16], customer service systems [17, 18], and autonomous driving [19] etc. In particular, VQA shows exceptional potential in the field of pathological diagnosis and health inquiry [20, 7]. Current medical VQA benchmarks include PMC-VQA [7], SLAKE [11], Path-VQA [8], and VQA-RAD [30]. However, these datasets are focused on medical diagnostics.

In agriculture, existing popular datasets like PlantVillage [4], PlantDoc [21], and PlantSeg [22] focus on classification or segmentation tasks. While they support disease detection, they do not enable interactive reasoning through question–answer formats. Recent systems like AgroGPT [23], LLaVa-PlantDiag [24] incorporate VQA models with the PlantVillage dataset. But these resources use generalized large language models for text generation and lack rigorous expert verification.

To address these gaps, we introduce PlantVillageVQA, a domain-specific visual question answering dataset for plant disease diagnosis. Built on 55,448 PlantVillage images, it contains 193,609 question–answer pairs covering nine distinct question categories under three levels of cognitive complexity. Each question was naturally phrased and tailored through expert review. Our methodology combines automated template-based QA generation with multistage linguistic re-engineering and iterative botanist review to ensure clinical accuracy and domain relevance. We further benchmark the dataset using CLIP [26], LXMERT [27], and FLAVA [29] to illustrate its utility for advancing multimodal agricultural research. To our knowledge PlantVillageVQA is the first multitmodal dataset created under extensive guidance by domain experts for the sole purpose of model training and evaluation.

Overall, our primary contributions include:

1. **PlantVillageVQA Dataset:** We introduce a domain-specific VQA dataset consisting of 193,609 expert-verified QA pairs across 55,448 plant images, covering 14 crop species and 38 disease classes.
2. **Question Taxonomy and Scope:** We design a diverse question set spanning nine categories such as basic identification to causal and counterfactual reasoning, enabling multi-level visual understanding.
3. **Two-Stage QA Generation Pipeline:** We implement an automated template-based QA synthesis pipeline. This was followed by a linguistically guided re-engineering phase to ensure semantic variety and answer diversity.
4. **Domain Expert Review:** The entire dataset underwent two phases of domain expert validation. This ensured clinical accuracy and relevancy to the field.

These contributions make PlantVillageVQA a useful and reliable dataset for research in plant disease diagnosis using visual question answering.

2 Methods

The PlantVillageVQA dataset was developed through a two-level data generation process: **Programming Based Data Generation, Data Re-engineering and Refinement** and a multilevel correction pipeline in the validation stage. Figure 1 shows a schematic description of the overall methodology in developing PlantVillageVQA.

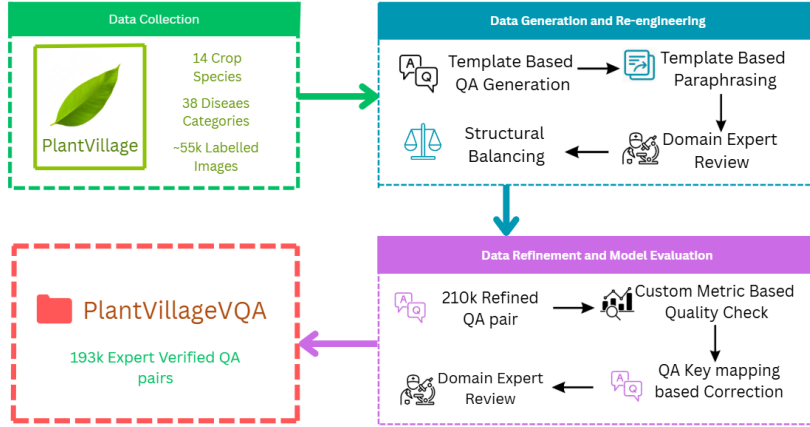


Figure 1: Overall Methodology of PlantVillageVQA creation

2.1 Programming Based Data Generation from PlantVillage

PlantVillage [28] is an open-access repository comprising 55,448 images of plant leaves of various conditions across 38 categories. A key feature here is the dataset’s organized hierarchical structure. Each directory in the dataset includes the labels for an image’s species and health status. For example a directory named **Tomato.Late_blight** contains only images of tomato leaves with late blight disease symptoms. Similarly, a directory labeled **Apple.Healthy** is comprised only healthy apple leaves images. We utilized this meticulous structure to automatically generate our VQA dataset as shown in Figure 2.

We proposed nine distinct categories each with multiple questions. These questions targeted specific criteria and required in-depth comprehension from the test models. These categories ranged from straightforward questions like **Plant Species Identification** to more complex inquiries like **Counterfactual Reasoning** and analytical tasks such as **Visual Attribute Grounding**. We avoided freeform generation with large language models during text generation. Each question type was formulated and refined based on feedback from three experienced graduate students from the Department of Botany, University of Dhaka. This design minimized unsupported claims and typical VLM/GPT hallucinations. It also made correction verifiable and auditable. The nine question category can be grouped into three levels of cognitive complexity. Each level and it’s question categories are shown in Table 1, Table 2 and Table 3 respectively.

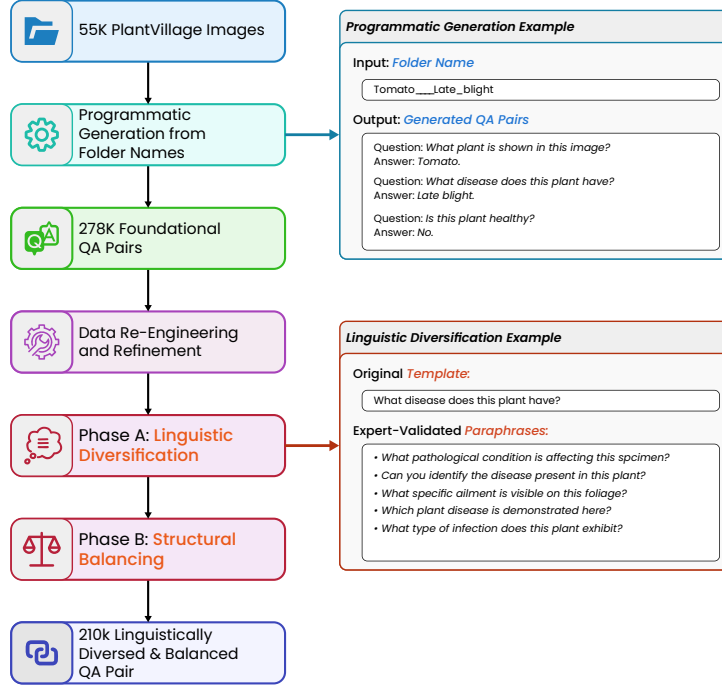


Figure 2: PlantVillageVQA Proposed Pipeline

Level 1: Foundational Perception and Identification

This level focuses on the model’s ability to identify elements in the image. These questions are direct and typically have factual, closed-set answers.

Question Category	Purpose	Template Design
1. Existence & Sanity Check	This category was designed to test to filter out irrelevant or background images. The templates were simple, direct, and binary.	"What is the primary subject of this image?"
2. Plant Species Identification	For species specific diseases, identifying the plant host is necessary. This category tests the model’s fine-grained visual categorization capabilities. Templates were designed by extracting crop name from folder name. Both positive and negative examples were generated to create a robust classification task.	Is this a [Crop] leaf?
3. General Health Assessment	This category tests the model’s ability to make a high-level, binary judgment of the plant’s overall health status. Templates were straightforward binary queries based on the "Disease" part of the folder name.	Is the plant in this image healthy?

Table 1: Question Categories for Foundational Perception

Level 2: Detailed Analysis and Verification

This is one of the most critical categories for diagnostics. It tests the model’s ability to detect, often subtle, visual symptoms mentioned in a textual phrase.

Question Category	Purpose	Template Design
1. Visual Attribute Grounding	Templates were created to describe the canonical visual symptoms for each of the 38 disease conditions, based on established phytopathology literature [31, 32].	Does the leaf exhibit dark, concentric ‘bullseye’ rings?
2. Detailed Verification	This category combines species identification and disease identification into a single, compositional query. Templates explicitly mentioned both the crop and the disease, creating highly specific binary questions.	Is this [Crop] leaf infected with [Disease_Name]?

Table 2: Question Categories for Detailed Analysis

Level 3: Higher-Order Reasoning and Inference

This level required the model to synthesize information, infer relationships, and reason about hypothetical scenarios.

Question Category	Purpose	Template Design
1. Specific Disease Identification	This is the direct, open-ended diagnostic task. Unlike verification, the model is not given a candidate disease; it must recall and provide the correct one from its knowledge base. The questions were phrased as direct requests for a diagnosis.	Please provide a diagnosis for the condition shown.
2. Comprehensive Description	This tests the model’s ability to generate a holistic, summary descriptions. Templates were open-ended prompts for descriptive text.	Provide a full description of the plant and its condition.
3. Causal Reasoning	This probes whether the model can infer the causal agent responsible for the visible symptoms. The questions were designed to ask for the cause, pathogen, or reason.	What is the cause of the unhealthy appearance of this leaf?
4. Counterfactual Reasoning	This task tests the model’s ability to reason about a hypothetical state that is contrary to the visual evidence. The templates posed hypothetical scenarios.	If this plant were healthy, what visual features would be different?

Table 3: Question Categories for Higher-Order Reasoning and Inference

For each image, our programming script extracted the crop and disease labels directly from the parent directory name and placed them in the aforementioned question templates. This

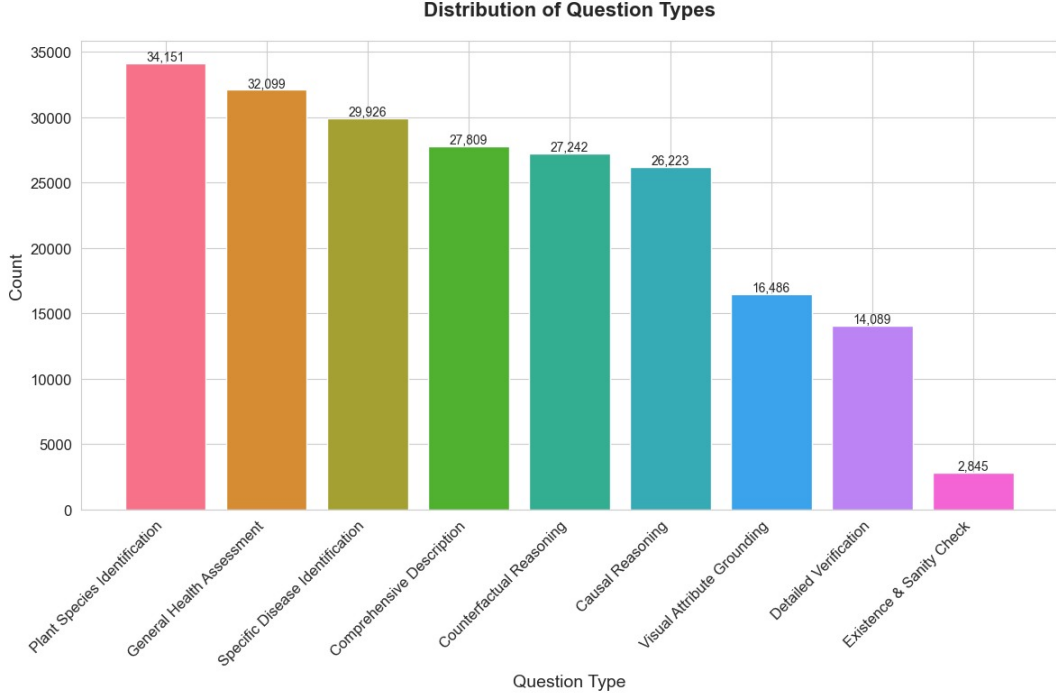


Figure 3: Question Frequency by Type

approach allowed us to generate a large number of questions in a very short time. In fact, by the end of the first stage we had created 278,255 Question-Answer (QA) pairs. Figure 3 illustrates how generated questions were distributed among different categories.

2.2 Data Refinement and Re-engineering

Following the completion of data generation, we conducted a thorough evaluation of our resulting database. Here, we focused on the vocabulary and semantic diversity of the generated QA pairs to assess how expressive our dataset is.

Most frequent questions	Count
Provide a full description of the plant and its condition.	27809
If this plant were healthy, what visual features would be different?	25152
Please provide a diagnosis for the condition shown	15004
What pathogen is responsible for the symptoms displayed here?	13118
Does this leaf appear to be free of disease?	10842

Table 4: Most Repeated Question Count

We searched for the top 5 repetitive questions and answers. The result is shown in Table 4. It can be seen that all top 5 questions were repeated over 10,000 times and counted 151,952 in total. This suggested a very rigid dataset, lacking nuance and diversity.

For vocabulary, we examined the most distinctive words used in both questions and answers.

We ignored structural elements such as articles and prepositions, focusing only on words with semantical influence. Our survey revealed the unique word-count across the model to be only 178 words.

2.2.1 Linguistic Diversification through Template-Based Paraphrasing

In order to introduce a more diverse vocabulary, we applied Template-Based Paraphrasing. We rephrased the same text multiple times while preserving its original meaning. For questions, we first identified the most frequent templates. Next we used a pool of 10-15 high-quality paraphraser to manually edit these templates.

Original Question	Paraphrase Pool
What disease does this [Crop] leaf have?	Identify the disease affecting this [Crop] leaf?
	Can you diagnose the ailment present on this [Crop] foliage?
	What pathological condition is evident on this [Crop] leaf?
	Which disease is indicated by the symptoms on this [Crop] leaf?
	Please specify the disease observed on this [Crop] leaf

Table 5: Question and its Paraphrase Pool

An example question and its paraphrase pool are shown in Table 5. At this stage, our team of specialists reviewed question variations from all nine categories to ensure scientific accuracy and consistency. We discarded all grammatically incorrect and excessively complex questions. Once validated, we replaced each question template with a randomly selected variation from its paraphrase pool. This process expanded our dataset vocabulary by 359%; making the dataset more communicative and accessible.

We measured the comprehensibility of each question type through the Flesch Reading Ease Score. This score is calculated by considering average sentence length and the average number of syllables per word. Higher score indicates better readability. Figure 4 shows this score varies across all question categories. This indicates the dataset covers both simple and complex inquiries.

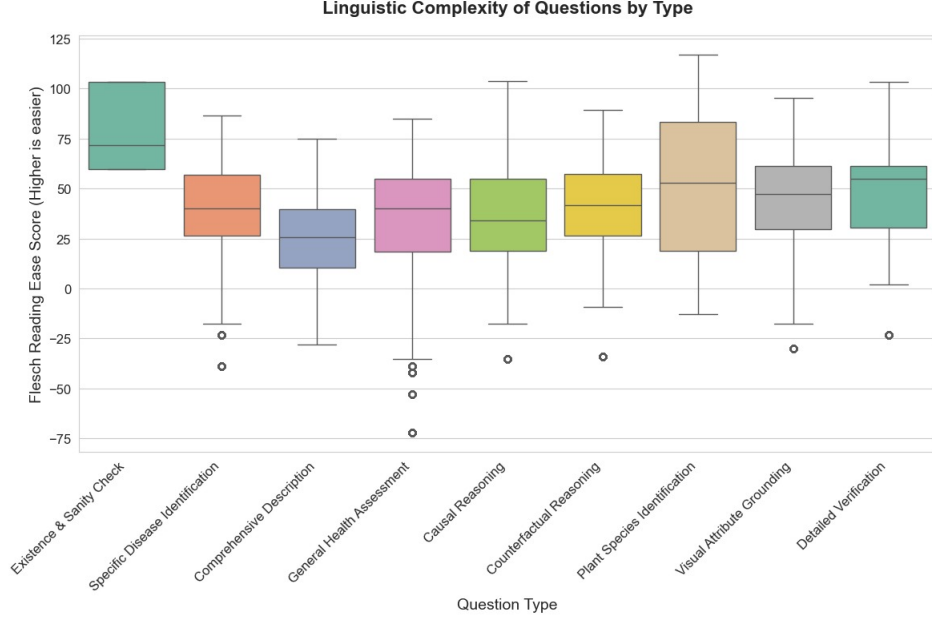


Figure 4: Question Linguistic Complexity

For answers, we employed a similar strategy. The dataset initially showed answer-side bias (for a given disease, a small fixed group of answers were provided). To counter this, we first identified the "answer bundles" for each major disease. Next we replaced them with a diverse pool of descriptive answers. For instance, all answers related to `Late_blight` (e.g, **"This is a leaf with Late_blight."**, **"The cause is Late_blight."**) were replaced by a random selection of more elaborative answers like: **"The large, dark, water-soaked lesions are a key sign of Late Blight."** and **"This is a classic presentation of Late Blight, caused by *Phytophthora infestans*."**.

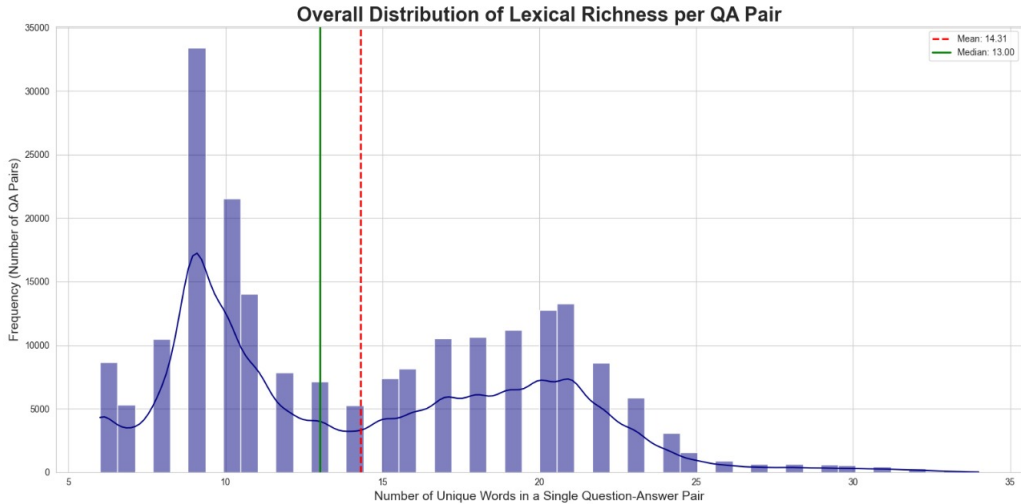


Figure 5: Lexical richness in QA pairs through re-engineering

The number of unique word-count per QA pair was calculated through the Lexical Richness Score. Figure 5 shows a mean count of 14.31 unique words per QA, meaning each pair uses approximately 14 words. This suggests the generated data has a rich range of vocabulary while

maintaining clarity.

Metric	Initial Dataset	Final Dataset	Increase	% Increase
Question Vocabulary	178	818	640	359.55%
Answer Vocabulary	148	201	53	35.81%

Table 6: Vocabulary Growth in Question and Answer Sets

We repeated this process iteratively until the number of high-frequency question and answers had reduced to close to 1000 words. As a result the number of unique word in questions and answers increased as shown in Table 6.

2.2.2 Targeted Stratified Undersampling for Structural Balance

Next we investigated if each question category was structurally balanced. We discovered that primarily three types of questions: **Visual Attribute Grounding**, **Detailed Verification** and **General Health Assessment** contributed to the overall imbalance.

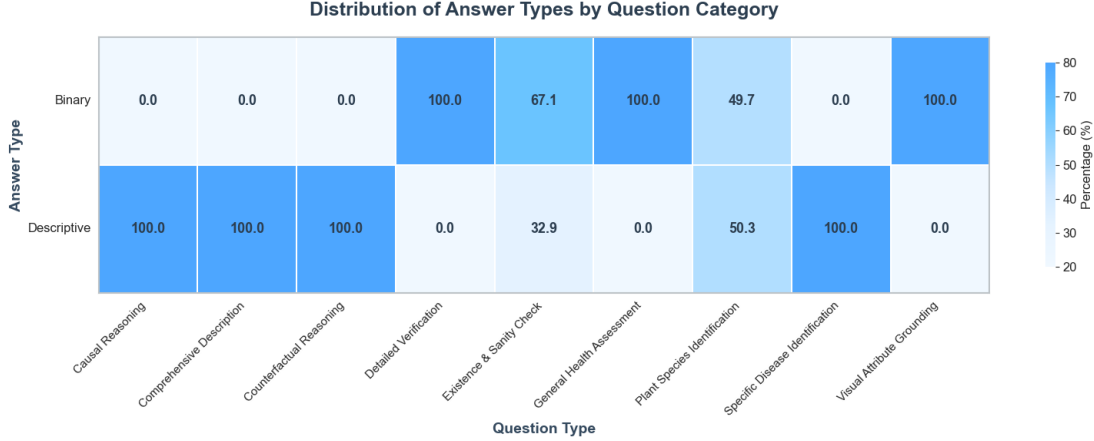


Figure 6: Heatmap of Answer type by Question Type

Figure 6 shows the heatmap of the binary and descriptive answers per question category. It can be seen that all three aforementioned categories have 100% binary answers. Further analysis showed that 78.6% of these questions were skewed toward negative answers. To solve this issue, we employed targeted stratified undersampling. Instead of undersampling the entire dataset, we first extracted the QA pairs only from the two problematic question types. There, we retained 100% of the "Yes" answer pairs and randomly reduced the "No" answer pairs to create a balanced 40/60 binary ratio. The 40/60 ration was a heuristic choice, employed specifically to preserve most of the dataset while also reducing the structural imbalance. Finally we merged the reduced QA bundles back into the original dataset. This resulted in an improved and balanced structure: the overall binary answer ratio being 59.4%.

These two re-engineering phases resulted in the final PlantvillageVQA dataset with 193,609 high-quality QA pairs. An image template with associated QA is shown in Figure 7.

Example from PlantVillage-VQA Image "image_000884.JPG"

■ Comprehensive Description

Q: Report all observable characteristics of this specimen's existing condition.
A: The plant shows no symptoms of pathology.

■ Counterfactual Reasoning

Q: What symptoms might appear if this Tomato were infected with Late_blight?
A: The plant is suffering from a Late Blight infection.

■ General Health Assessment

Q: Does this leaf display symptoms indicative of infection?
A: No

■ Plant Species Identification

Q: Is this a Corn leaf?
A: No

■ Specific Disease Identification

Q: Can you identify any pathological conditions affecting this tomato foliage?
A: This is a healthy plant leaf.



Figure 7: Image and its associated Questions

3 Data Records

The release comprises one image directory and two parallel annotation files. The `images/` directory contains **55,448** JPEG files named in a flat sequence: `images/image_000001.jpg`, `images/image_000002.jpg`, ... No crop or disease subfolders are used. Each annotation record points to a file in this directory via a relative path.

Two annotation files document **193,609** question-answer (QA) pairs: `PlantVillageVQA.csv` and `PlantVillageVQA.json`. Both contain identical content and schema; choose either format for downstream use.

`PlantVillageVQA.csv` is UTF-8, comma-separated, with the following columns:

- `image_id`: unique identifier of the image.
- `question_type`: one of the nine category labels.
- `question`: natural-language prompt.
- `answer`: reference answer of the question
- `image_path`: relative path, e.g., `images/image_000123.jpg`.
- `split`: dataset partition tag: `train` or `test`.

We have deposited the complete dataset on *Figshare*. It can be cited through the permanent identifier DOI: 10.6084/m9.figshare.29663456. During peer review, the private access link is <https://figshare.com/s/111f97bb350d52f10218>.

We will also provide public copies for convenience after acceptance of the manuscript: the dataset will be available on Hugging Face Hub (<https://huggingface.co/datasets/SyedNazmusSakib/PlantVillageVQA>), Kaggle (<https://www.kaggle.com/datasets/syednazmussakib/plantvillagevqa>), and the full codebase will be hosted on GitHub (<https://github.com/SyedNazmusSakib-SNS/PlantVillageVQA>).

4 Technical Validation

To ensure dataset quality, we combined expert validation with automated quality evaluation. The following sections outline each step of the validation process.

4.1 Domain Expert Review: Phase One

For expert evaluation, we created a web interface hosting our entire dataset. This interface allowed a team of experienced botanists to review question-answer relevancy and identify related issues through a submission form. Example page from the website is shown in Figure 8.

Image & Question-Answer Pair  CAUSAL REASONING What is the cause of the unhealthy appearance of this Potato leaf? ANSWER This is Late Blight, caused by the oomycete <i>Phytophthora infestans</i>.	Pathologist Feedback 1. How relevant and accurate is this Question-Answer pair to the provided image? ☐ Excellent: Accurate, relevant, high-quality ☐ Good: Mostly accurate, minor issues ☐ Okay: Understandable but noticeable flaws ☐ Poor: Largely irrelevant or incorrect ☐ Unsalvageable: Completely wrong 2. What is the primary issue with this QA pair? (Check all that apply) ☐ Factual Error in Answer: Incorrect disease/plant information ☐ Factual Error in Question: Incorrect assumption ☐ Vague/Ambiguous: Poorly phrased and unclear ☐ Symptom Not Visible: Discusses invisible symptom ☐ Logical Mismatch: Answer doesn't match question	Image & Question-Answer Pair  SPECIFIC DISEASE IDENTIFICATION What type of pathological condition is evident on this tomato foliage? ANSWER This is damage from Spider Mites.	Pathologist Feedback 1. How relevant and accurate is this Question-Answer pair to the provided image? ☐ Excellent: Accurate, relevant, high-quality ☐ Good: Mostly accurate, minor issues ☐ Okay: Understandable but noticeable flaws ☐ Poor: Largely irrelevant or incorrect ☐ Unsalvageable: Completely wrong 2. What is the primary issue with this QA pair? (Check all that apply) ☐ Factual Error in Answer: Incorrect disease/plant information ☐ Factual Error in Question: Incorrect assumption ☐ Vague/Ambiguous: Poorly phrased and unclear ☐ Symptom Not Visible: Discusses invisible symptoms ☐ Logical Mismatch: Answer doesn't match question
Image & Question-Answer Pair  COUNTERFACTUAL REASONING What visual features would return upon this specimen's complete health restoration? ANSWER If the plant were healthy, the symptoms of disease would be absent.	Pathologist Feedback 1. How relevant and accurate is this Question-Answer pair to the provided image? ☐ Excellent: Accurate, relevant, high-quality ☐ Good: Mostly accurate, minor issues ☐ Okay: Understandable but noticeable flaws ☐ Poor: Largely irrelevant or incorrect ☐ Unsalvageable: Completely wrong 2. What is the primary issue with this QA pair? (Check all that apply) ☐ Factual Error in Answer: Incorrect disease/plant information ☐ Factual Error in Question: Incorrect assumption ☐ Vague/Ambiguous: Poorly phrased and unclear ☐ Symptom Not Visible: Discusses invisible symptom ☐ Logical Mismatch: Answer doesn't match question	Image & Question-Answer Pair  CAUSAL REASONING Which pathological processes are contributing to this tomato leaf's condition? ANSWER This is Tomato Mosaic Virus (ToMV).	Pathologist Feedback 1. How relevant and accurate is this Question-Answer pair to the provided image? ☐ Excellent: Accurate, relevant, high-quality ☐ Good: Mostly accurate, minor issues ☐ Okay: Understandable but noticeable flaws ☐ Poor: Largely irrelevant or incorrect ☐ Unsalvageable: Completely wrong 2. What is the primary issue with this QA pair? (Check all that apply) ☐ Factual Error in Answer: Incorrect disease/plant information ☐ Factual Error in Question: Incorrect assumption ☐ Vague/Ambiguous: Poorly phrased and unclear ☐ Symptom Not Visible: Discusses invisible symptoms ☐ Logical Mismatch: Answer doesn't match question

Figure 8: Phase One Expert Review Form

Initial Specialist Feedback showed satisfactory performance across eight question categories. But some associated answers relied heavily on generic fallback; they contained no correlation with the questions. This was prevalent in the Counterfactual reasoning questions. While the questions themselves were correctly posed hypothetical scenarios (e.g., "What visual features would be different if this plant were healthy?") Their answers were occasionally paired with simple diagnostic statements (e.g., A: "The diagnosis is Tomato Yellow Leaf Curl Virus.").

We identified this problem to be caused by poor specificity in this category specific answer generation. To address this problem, we performed Hierarchical Correction Pipeline to all 27,242 counterfactual QA pairs.

4.1.1 Implementation of the Hierarchical Correction Pipeline

First, we leveraged the dataset’s own internal knowledge to mine all 16,486 questions from the Visual Attribute Grounding category. These contained expert-phrased, canonical descriptions of visual symptoms (e.g., ”Does the leaf exhibit dark, concentric ’bullseye’ rings?”). We programmatically extracted these descriptions and mapped them to key symptom words (e.g., ’bullseye’ → ”dark, concentric ’bullseye’ rings”), creating a **canonical_phrase map**.

To enhance counterfactual specificity we adopted a simple, defensible rule: fix when verifiable; delete when not. Any QA pair was verifiable when it contained all three of the following: (i) valid **crop_disease** provenance, (ii) a **disease_keyword** that maps to a condition, and (iii) an expert symptom phrase for that condition. If any element above was missing (e.g., generic answer like ”The plant is healthy.”), we deleted the QA pair. We then regenerated the valid counterfactual answers with the template: ”A healthy leaf would not show ... [canonical symptom] ...”. We refrained from using LLM models for question refinement to eliminate hallucination risk and preserve traceability. The numerical effect of the process on the counterfactual question category is shown in Table 7.

Initial Pool	27,242 QA Pairs
Corrected (Regenerated Pool)	9,981 QA Pairs (36%)
Deleted (Non Verifiable)	17,261 QA Pairs (63.7%)

Table 7: Quantitative Outcome of Hierarchical Correction

This ensured every surviving counterfactual answer was logically responsive, symptom-grounded, and reproducible from code and maps.

4.1.2 Impact of Logical Correction

The Hierarchical Correction reduced the models’ reliance on the generic fallback, indicating a significant improvement in the counterfactual question category. Table 8 provides a quantitative comparison of the dataset before and after final refinement.

Metric	Before Expert Feedback (v4.4)	After Expert Feedback (v4.5)	Improvement
Total Generic ”Fallback” Answers	14,991	1,496	-13,495
Generic Answers as % of All Counterfactuals	55.03%	5.49%	A 90% reduction in generic answers
Frequency of a Single Generic Template	3,106	395	An 87% reduction in repetition

Table 8: Comparison of Metrics Before and After Specialist Feedback and Correction

4.2 Automated Quality Evaluation and Outlier Detection

After initial botanist review and correction, we opted for another thorough analysis of the entire database. We developed an automated pipeline to highlight suspicious QA pairs for expert re-evaluation. This process quantified abstract notions of quality, such as "simplicity", "vagueness" and "relevance" using established techniques from natural language processing and information theory.

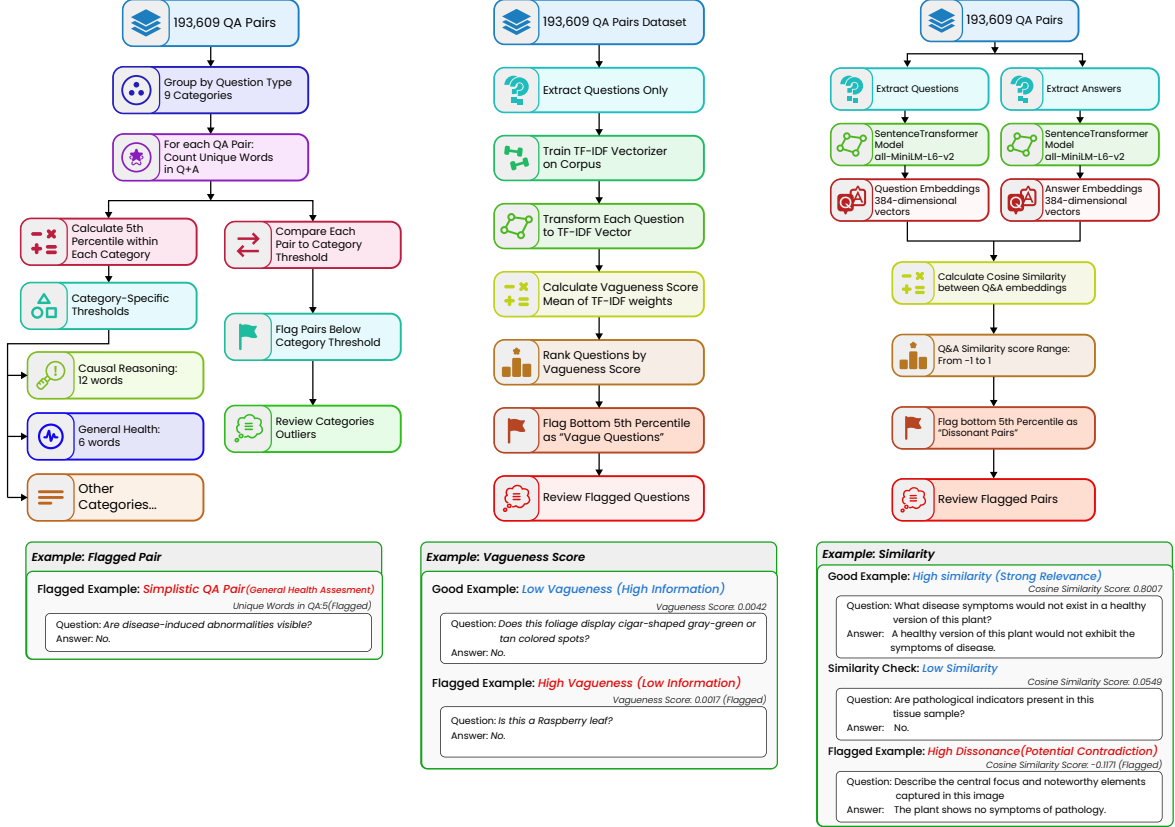


Figure 9: Working flowcharts of Automated Quality Evaluation and Outlier Detection: Relative Simplicity (left), Vagueness Score Assessment (center) and Semantic Dissonance (right).

Three distinct analyses were performed on all 193,609 QA pairs as shown in Figure 9. These metrics do not indicate inaccuracy in questions, but low conversation ability. For example the question, "Is this a Raspberry leaf?" is short but not vague in intent. These analyses allowed us to conduct a rigorous recheck of the entire database for communicative value.

Relative Simplicity: We assessed relative simplicity by counting unique words per QA pairs and comparing them with the mean unique word counts of their question category. Question below the 5 percentile threshold for each category (e.g., Casual Reasoning question with 5 unique words) were flagged as outliers. The flowchart in the left of Figure 9 shows this assessment.

Vagueness Score: We hypothesized that vague or low value questions heavily rely on common words (e.g., "what," "is," "leaf"), lacking specific keywords. To quantify this, we calculated vagueness score using TF-IDF weight of every word in a question. Lower score indicated less informative content and the bottom 5 percentile of such question were flagged. The center flowchart in Figure 9 shows the process.

Semantic Similarity: To detect semantic similarity between QA pairs, we used sentence-transformer embeddings (all-MiniLM-L6-v2) and computed QA similarity via cosine similarity. Lower scores identified less question-answer relevancy. This process is shown in the center flowchart of Figure 9. A total number of 25,418 QA pairs were flagged by the outlier detection pipeline.

4.3 Domain Expert Review: Phase Two

In phase two, we focused only on the 25,418 QA pairs flagged by the automated outlier detection pipeline. We created a second web interface that showed samples from this compilation. This form allowed reviewers to either keep or discard the QA pairs. An example page of the form is shown in Figure 10.

Figure 10: Phase Two Expert Review Form

During this final inspection, our team of botanists went through a random selection of 2837 these forms, out of which only 91 were discarded. This indicated a 97.57% correctness, proving the Dataset to be ready for model application and benchmarking.

4.4 Model Applications and Benchmarks

To validate the integrity of the re-engineered PlantVillageVQA dataset, we conducted a benchmarking study using three state-of-the-art vision-language models: CLIP [26], LXMERT [27], FLAVA [29].

Our objective was twofold: first, to demonstrate that the dataset contains learnable patterns enabling models to outperform random chance; and second, to show that the dataset is sufficiently complex to challenge model performance. To provide a holistic evaluation, we chose the metrics shown in Table 9.

Metric	General Description	Usage in PlantVQA
Accuracy	All "true positive" responses divided by total predictions; calculates the percentage of exact matches.	Used for binary (Yes/No) questions and single-word answers. Relevant to: Existence & Sanity Check, Specific Disease Identification, General Health Assessment.
BLEU (Bilingual Evaluation Understudy)	Measures the precision of n-grams (contiguous word sequences); it is used to evaluate the quality of machine generated text.	Applied to categories with descriptive answers: Comprehensive Description, Visual Attribute Grounding, Causal Reasoning, Counterfactual Reasoning.
METEOR (Metric for Evaluation of Translation with Explicit Ordering)	Aligns generated answers with references using both exact and semantic similarity: synonyms and word order.	Used in: Comprehensive Description, Causal Reasoning.
ROUGE (Recall-Oriented Understudy for Gisting Evaluation)	Used to evaluate machine generated summaries/translations with human references by overlapping word sequence.	Useful for: Comprehensive Description, Visual Attribute Grounding; captures completeness of longer responses.

Table 9: Evaluation Metrics and Their Usage in the PlantVillageVQA Dataset

4.4.1 Results

Model	Accuracy	BLEU	METEOR	ROUGE-1	ROUGE-L
Flava	0.3432	0.0909	0.2066	0.3826	0.3764
CLIP	0.6148	0.2452	0.4476	0.7323	0.7153
LXMERT	0.6034	0.2382	0.4393	0.7219	0.7061

Table 10: Flava, CLIP, and LXMERT Models' Performance across Metric

Table 10 shows that CLIP and LXMERT achieved the highest performance across all metrics, with CLIP attaining a top accuracy of 0.6148. This score is substantially better than random chance. It demonstrates that the dataset contains clear, learnable visual and linguistic patterns that a capable model can exploit. Moreover, the model's lack of perfect accuracy confirms our efforts to linguistically diversify have resulted in a dataset that challenges simple solutions.

The text generation metric like the ROUGE-1 scores of over 0.72 for LXMERT and CLIP suggest that the models are effective at identifying key terms present in the reference answers. However, metrics like BLEU (0.24) and METEOR (0.43) are more modest. This indicates that while models can often generate the correct keywords, they struggle with the semantic variety of the full descriptive answers we created in our refinement phase.

The performance gap between FLAVA and the other two models, may be attributed to the nature of pre-training. CLIP's pre-training on extremely diverse, web-scale data likely provided it with a more robust and generalizable visual representation system. This proved

highly effective for the fine-grained distinctions required in plant pathology.

Figure 11 shows accuracy of each model by question category. It can be seen that all three models perform relatively well on binary and identification related questions, (e.g., Plant species identification, Detailed Verification, Existence & Sanity Check). But each model struggled with questions requiring detailed reasoning and deep domain understanding (e.g., Casual Reasoning, Counterfactual Reasoning, Detailed verification, Specific Disease identification).

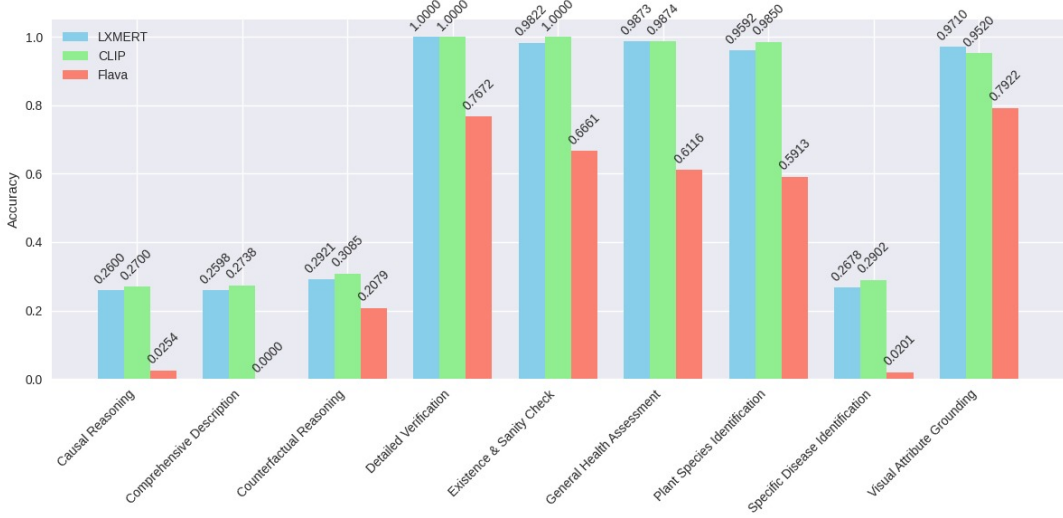


Figure 11: Model Performance Across Metrics

In conclusion, this model-based validation demonstrates that the PlantVillageVQA dataset is a valid and effective resource for training complex VQA models. It also proves that the dataset is a challenging benchmark, with significant headroom for improvement, even for state-of-the-art architectures. This positions PlantVillageVQA as a valuable tool for driving future research into more capable, domain-specialized systems for agricultural diagnostics.

Code Availability

For peer-review purposes the full codebase can be downloaded from the following Google Drive folder: <https://drive.google.com/drive/folders/1hwx-Q20sShYT8SX8IRyWHIzoRofz60ms?usp=sharing>. The complete dataset generation process and how to read and visualize the dataset are provided in the GitHub repository: <https://github.com/SyedNazmusSakib-SNS/PlantVillageVQA>. The repository is currently private and will be made publicly accessible upon acceptance of the article.

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Author Contributions

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Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Md Abdullahil Baki Bhuiyan, Hasan Muhammad Abdullah, Shifat E Arman, Sayed Saminur Rahman, and Kaies Al Mahmud. BananaSqueezeNet: A very fast, lightweight convolutional neural network for the diagnosis of three prominent banana leaf diseases. *Smart Agricultural Technology*, 4:100214, 2023.
- [2] Md Arban Hossain, Saadman Sakib, Hasan Muhammad Abdullah, and Shifat E Arman. Deep learning for mango leaf disease identification: A vision transformer perspective. *Heliyon*, 10(17), 2024.
- [3] Serge Savary, Laetitia Willocquet, Sarah Jane Pethybridge, Paul Esker, Neil McRoberts, and Andy Nelson. The global burden of pathogens and pests on major food crops. *Nature Ecology & Evolution*, 3(3):430–439, 2019.
- [4] Sharada P. Mohanty, David P. Hughes, and Marcel Salathé. Using deep learning for image-based plant disease detection. *Frontiers in Plant Science*, 7:215232, 2016.
- [5] Konstantinos P. Ferentinos. Deep learning models for plant disease detection and diagnosis. *Computers and Electronics in Agriculture*, 145:311–318, 2018.
- [6] Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C. Lawrence Zitnick, and Devi Parikh. VQA: Visual question answering. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2425–2433, 2015.
- [7] Xiaoman Zhang, Chaoyi Wu, Ziheng Zhao, Weixiong Lin, Ya Zhang, Yanfeng Wang, and Weidi Xie. PMC-VQA: Visual instruction tuning for medical visual question answering. *arXiv preprint arXiv:2305.10415*, 2023.

- [8] Xuehai He, Yichen Zhang, Luntian Mou, Eric Xing, and Pengtao Xie. PathVQA: 30000+ questions for medical visual question answering. *arXiv preprint arXiv:2003.10286*, 2020.
- [9] Zichao Yang, Xiaodong He, Jianfeng Gao, Li Deng, and Alex Smola. Stacked attention networks for image question answering. *arXiv preprint arXiv:1511.02274*, 2015.
- [10] Yash Goyal, Tushar Khot, Daniel Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in Visual Question Answering. *arXiv preprint arXiv:1612.00837*, 2017.
- [11] Bo Liu, Li-Ming Zhan, Li Xu, Lin Ma, Yan Yang, and Xiao-Ming Wu. SLAKE: A semantically-labeled knowledge-enhanced dataset for medical visual question answering. In *2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI)*, pages 1650–1654. IEEE, 2021.
- [12] Soravit Changpinyo, Piyush Sharma, Nan Ding, and Radu Soricut. Conceptual 12M: Pushing web-scale image-text pre-training to recognize long-tail visual concepts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3558–3568, 2021.
- [13] Hedi Ben-Younes, Rémi Cadene, Matthieu Cord, and Nicolas Thome. MUTAN: Multi-modal Tucker Fusion for Visual Question Answering. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2612–2620, 2017.
- [14] Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6700–6709, 2019.
- [15] S. Banerjee and S. Bhattacharya. VQA for Education: A Survey of Recent Trends and Future Directions. *arXiv preprint arXiv:2210.12345*, 2022.
- [16] Y. Huang and Z. Wang. AI Tutors in the Classroom: A VQA-based Approach. *Journal of Educational Technology*, 48(2):112–128, 2022.
- [17] A. Jain and R. Gupta. A VQA-based System for Customer Support in E-commerce. In *Proceedings of the International Conference on Information Systems (ICIS)*, pages 1–10, 2021.
- [18] Q. Zhu and X. Li. Enhancing E-commerce Experience with Visual Question Answering. *Journal of Retail and E-commerce*, 12(3):45–62, 2023.
- [19] Ana-Maria Marcu, Long Chen, Jan Hünemann, Alice Karnsund, Benoit Hanotte, Prajwal Chidananda, Saurabh Nair, Vijay Badrinarayanan, Alex Kendall, Jamie Shotton, and others. LingoQA: Visual question answering for autonomous driving. In *European Conference on Computer Vision (ECCV)*, pages 252–269. Springer, 2024.
- [20] K. Yan and Y. Wang. RadVQA: A Visual Question Answering Benchmark for Radiology. *Journal of Medical Imaging*, 10(1):014501, 2023.

- [21] Davinder Singh, Naman Jain, Pranjali Jain, Pratik Kayal, Sudhakar Kumawat, and Nipun Batra. PlantDoc: A dataset for visual plant disease detection. In *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD*, pages 249–253, 2020.
- [22] Adrian Wolny, Lorenzo Cerrone, Athul Vijayan, Rachele Tofanelli, Amaya Vilches Barro, Marion Louveaux, Christian Wenzl, Sören Strauss, David Wilson-Sánchez, Rena Lymbouridou, and others. Accurate and versatile 3D segmentation of plant tissues at cellular resolution. *eLife*, 9:e57613, 2020.
- [23] Muhammad Awais, Ali Husain Salem Abdulla Alharthi, Amandeep Kumar, Hisham Cholakkal, Rao Muhammad Anwer, and others. AgroGPT: Efficient agricultural vision-language model with expert tuning. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 5687–5696. IEEE, 2025.
- [24] Karun Sharma, Vidushee Vats, Abhinendra Singh, Rahul Sahani, Deepak Rai, and Ashok Sharma. LLaVA-PlantDiag: Integrating Large-scale Vision-Language Abilities for Conversational Plant Pathology Diagnosis. In *2024 International Joint Conference on Neural Networks (IJCNN)*, pages 1–7. IEEE, 2024.
- [25] Yunpeng Zhao, Shansong Wang, Qingtian Zeng, Weijian Ni, Hua Duan, Nengfu Xie, and Fengjin Xiao. Informed-Learning-Guided Visual Question Answering Model of Crop Disease. *Plant Phenomics*, 2024, 2024.
- [26] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, and others. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, pages 8748–8763. PMLR, 2021.
- [27] Hao Tan and Mohit Bansal. LXMERT: Learning cross-modality encoder representations from transformers. *arXiv preprint arXiv:1908.07490*, 2019.
- [28] David Hughes, Marcel Salathé, and others. An open access repository of images on plant health to enable the development of mobile disease diagnostics. *arXiv preprint arXiv:1511.08060*, 2015.
- [29] Amanpreet Singh, Ronghang Hu, Vedanuj Goswami, Guillaume Couairon, Wojciech Galuba, Marcus Rohrbach, and Douwe Kiela. FLAVA: A foundational language and vision alignment model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15638–15650, 2022.
- [30] Jason J. Lau, Soumya Gayen, Asma Ben Abacha, and Dina Demner-Fushman. A dataset of clinically generated visual questions and answers about radiology images. *Scientific Data*, 5(1):1–10, 2018.
- [31] George N. Agrios. *Plant Pathology*. Elsevier, 2005.
- [32] Richard N. Strange and Peter R. Scott. Plant disease: a threat to global food security. *Annual Review of Phytopathology*, 43(1):83–116, 2005.